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Tutorial 9

1.

Let X be the random variable of the total winnings of the player.

We have: random variable X takes the values $x = -\$1$ (player loses) and $x = \$499$ (player wins)

The probability of getting three 6s is $\frac{1}{6^3} = \frac{1}{216}$

$$\Rightarrow P(X=-1) = \frac{215}{216}$$

$$P(X=499) = \frac{1}{216}$$

The expected winning of this game:

$$E(X) = \sum_i p_i x_i$$

$$= -1 \cdot \frac{215}{216} + 499 \cdot \frac{1}{216}$$

$$\approx 1,31$$

My answer about whether I would play this game depends upon how many times I can play it

+ If I can play this game once, I will not play it. This is because the odd of winning is only $\frac{1}{216}$, which is very low.

+ If I can play the game many times, I will play it and the more times of playing I have, the more I am willing to pay. This is because the expected winning is positive, so in the long run I will benefit from the game.

2.

a) The expectation of X:

$$E(X) = -2 \cdot \frac{1}{3} + 1 \cdot \frac{1}{6} + 4 \cdot \frac{1}{3} + 6 \cdot \frac{1}{6}$$

$$= \frac{11}{6}$$

b) The variance of X:

$$Var(X) = E((X - E(X))^2)$$

$$= \sum_i p_i (x_i - E(X))^2$$

$$= \frac{1}{3} \cdot \left(-2 - \frac{11}{6}\right)^2 + \frac{1}{6} \cdot \left(1 - \frac{11}{6}\right)^2$$

$$+ \frac{1}{3} \cdot \left(4 - \frac{11}{6}\right)^2 + \frac{1}{6} \cdot \left(6 - \frac{11}{6}\right)^2$$

$$= \frac{341}{36}$$

c) The variance of X:

$$Var(X) = E(X^2) - (E(X))^2$$

$$= \sum_i p_i \cdot x_i^2 - (E(X))^2$$

$$= \frac{1}{3} \cdot (-2)^2 + \frac{1}{6} \cdot 1^2 + \frac{1}{3} \cdot 4^2 + \frac{1}{6} \cdot 6^2 - \left(\frac{11}{6}\right)^2$$

$$= \frac{341}{36}$$

3.

a) Let X be the event that I have a profit.

According to slide 7 of Lecture 9, we can calculate the profit for each cases and the probability, respectively.

Profit	x_i	-3	-2	-1	0	1	2
Probabilities	p_i	1/18	1/9	1/6	2/9	5/18	1/6

The expected profit ~~from each game~~ is:

$$E(X) = 2 \cdot \frac{1}{6} + 1 \cdot \frac{5}{8} + 0 \cdot \frac{2}{9} + (-1) \cdot \frac{1}{6} + (-2) \cdot \frac{1}{9} + (-3) \cdot \frac{1}{18}$$

$$= \frac{1}{18}$$

$$E(X^2) = \sum p_i \cdot x_i^2 = \frac{1}{6} \cdot 2^2 + \frac{5}{18} \cdot 1^2 + \frac{2}{9} \cdot 0 + \frac{1}{6} \cdot (-1)^2 + \frac{1}{9} \cdot (-2)^2 + \frac{1}{18} \cdot (-3)^2$$

$$= \frac{37}{18}$$

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The variance in my profit from each game:

$$Var(X) = E(X^2) - (E(X))^2$$

$$= \frac{37}{18} - \left(\frac{1}{18}\right)^2 = \frac{665}{324}$$

$$\approx 2,052$$

I prefer a small variance because it ensures me a stability by following explanation:

Although the expected value shows the average win and lose but the variance depicts the level of fluctuation around the expected value. That means if the variance is high, the result may be very different from the expected value and if the variance is low, the result will be ~~around~~ more around the expected value. So, in this excercise, we have positive expected winning and it is a useful information to choose the low variance over high variance,

b) The probability of rolling 3 = 0,2

The probability of rolling 1,2,3,4,5,6 = 0,16

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Let X be the event of expected difference.
We can calculate new probabilities of the positive difference between the score:

x_i	0	1	2	3	4	5
p_i	0,168	0,2816	0,2304	0,1664	0,1024	0,0512

The expectation of X:

$$E(X) = 0 \cdot 0{,}168 + 1 \cdot 0{,}2816 + 2 \cdot 0{,}2804 + 3 \cdot 0{,}1664 + 4 \cdot 0{,}1024 + 5 \cdot 0{,}0512 = 1{,}9072$$

So ~~expected~~ the profit increase beyond 6 cents per game because $2 - 1{,}9072 = 0{,}0928$ which is larger than 0,06.

No, it is not suprised at all. Because 3 is close to the expectation of the value of a fair die when rolled once. So when the likelihood of rolling a three increases, the variance ~~of one roll~~ will decrease, and the expected difference between two rolls will decrease as well.

4.

a) Because random variable X is in the interval [3;4], we have the below probability density function:

$$\int_3^4 f(x)\,dx = \int_3^4 \frac{A}{\sqrt{x}}\,dx = 1$$

$$\Rightarrow \int_3^4 f(x)\,dx = A.2\sqrt{4} - A.2.\sqrt{3} = 1$$

$$\Rightarrow 2A(2-\sqrt{3}) = 1$$

$$\Rightarrow A = \frac{1}{4-2\sqrt{3}} = \frac{2+\sqrt{3}}{2}.$$

b) Let A equals to $\frac{2+\sqrt{3}}{2}$

We have $f(x) = \frac{A}{\sqrt{x}}$ (for $3 \le x \le 4$)

$$F(x) = P(X \le x) = \int_3^x \frac{A}{\sqrt{y}}\,dy$$

$$= 2A \int_3^x \frac{1}{2\sqrt{y}}\,dy$$

$$= 2A.\sqrt{y}\Big|_3^x$$

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$$= 2A.(\sqrt{x} - \sqrt{3})$$

$$= 2A.\sqrt{x} - 2A\sqrt{3}$$

$$\Rightarrow F(x) = (2+\sqrt{3})\sqrt{x} - (3+2\sqrt{3})$$

c) The expected value of X is:

$$E(X) = \int_3^4 x.\frac{2+\sqrt{3}}{2\sqrt{x}}\,dx$$

$$= 3,488$$

d) The standard diviation of X is:

$$\sigma = \sqrt{\int_3^4 \frac{2+\sqrt{3}}{2\sqrt{x}}.\left(x - \int_3^4 x.\frac{2+\sqrt{3}}{2\sqrt{x}}\right)^2 dx}$$

$$= 0,2887$$

e) The median of X is:

$$F(x) = 0,5$$

$$\Leftrightarrow (2+\sqrt{3})\sqrt{x} - (3+2\sqrt{3}) = 0,5$$

$$\Leftrightarrow x = \frac{7+4\sqrt{3}}{4} \approx 3,482$$

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